

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. Find the inverse of the function $f(x) = x^2 + 2x + 1$ for $x \geq -1$. Show your work and explain your reasoning.

Q10. A company's profit P (in thousands of dollars) from an investment of x thousand dollars is given by the function $P(x) = 2x - x^2$. Find the inverse function and interpret its meaning in this context. Be sure to state any domain restrictions.

Chef's Tips

- Ensure students show their work for free response questions.
- Provide feedback on the correctness of the inverse functions.
- Check for domain restrictions on the inverse functions.